Fake News Detection

Objective : The objective of this assignment is to develop a Semantic Classification model. You will be using Word2Vec method to extract the semantic relations from the text and develop a basic understanding of how to train supervised models to categorise text based on its meaning, rather than just syntax. You will explore how this technique is used in situations where understanding textual meaning plays a critical role in making accurate and efficient decisions.

Business Objective : The spread of fake news has become a significant challenge in today's digital world. With the massive volume of news articles published daily, it's becoming harder to distinguish between credible and misleading information. This creates a need for systems that can automatically classify news articles as true or fake, helping to reduce misinformation and protect public trust.

In this assignment, you will develop a Semantic Classification model that uses the Word2Vec method to detect recurring patterns and themes in news articles. Using supervised learning models, the goal is to build a system that classifies news articles as either fake or true.

1. Data Preparation

1.0 Data Understanding

Data Understanding is the process of exploring, inspecting, and familiarizing yourself with the raw data so you can make informed decisions during data preparation and modeling.

Using few tools like head(), info(), describe() we can try to understand the raw data.

```
# Print the column details for True News DataFrame
df_true_wc.info()
```

✓ 0.0s

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 21417 entries, 0 to 21416
Data columns (total 3 columns):
 #   Column  Non-Null Count  Dtype
---  ------  --------------  -----
 0   title   21417 non-null  object
 1   text    21417 non-null  object
 2   date    21417 non-null  object
dtypes: object(3)
memory usage: 502.1+ KB
```

```
# Print the column details for Fake News Dataframe
df_fake_wc.info()
```

✓ 0.0s

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 23523 entries, 0 to 23522
Data columns (total 3 columns):
 #   Column  Non-Null Count  Dtype
---  ------  --------------  -----
 0   title   23502 non-null  object
 1   text    23502 non-null  object
 2   date    23481 non-null  object
dtypes: object(3)
memory usage: 551.4+ KB
```

```
# Print the column names of both DataFrames
print("Fake News Columns:", df_fake_wc.columns)
print("True News Columns:",df_true_wc.columns)
```

✓ 0.0s

```
Fake News Columns: Index(['title', 'text', 'date'], dtype='object')
True News Columns: Index(['title', 'text', 'date'], dtype='object')
```

1.1 Add new column

As we have two data sets, one for fake news and one for true news, we can add a new column called news label and assign value 1 for true and 0 for fake news. So that once both the datasets are merged this column will help us to identify the true/fake news.

```
# Add a new column 'news_label' to the true news DataFrame and assign the label "1" to indicate that these news are true
df_true_wc['news_label'] = 1
# Add a new column 'news_label' to the fake news DataFrame and assign the label "0" to indicate that these news are fake
df_fake_wc['news_label'] = 0
```

✓ 0.0s

1.2 Merge DataFrames

Combine the true and fake news DataFrames into a single DataFrame.

```
# Combine the true and fake news DataFrames into a single DataFrame
df_comb = pd.concat([df_true_wc, df_fake_wc])
```
✓ 0.0s

1.3 Handle the null values

First we need to check if any null values present in the data frame and then drop the null values if required.

```
# Check Presence of Null Values
df_comb.isnull().sum()
```
✓ 0.0s

```
title         21
text          21
date          42
news_label     0
dtype: int64
```

```
# Handle Rows with Null Values
df_comb = df_comb.dropna()
df_comb.isnull().sum()
```
✓ 0.0s

```
title         0
text          0
date          0
news_label    0
dtype: int64
```

1.4 Merge the relevant columns and drop the rest from the DataFrame

In this step we can combine the relevant columns into one and drop rest of the columns. Here we will merge the title and text column and create a new column called news_text and drop these two columns (title and text).

```
# Combine the relevant columns into a new column 'news_text' by joining their values with a space

df_comb['news_text'] = df_comb['title'] + ' ' + df_comb['text']

# Drop the irrelevant columns from the DataFrame as they are no longer needed
df_comb = df_comb.drop(['title', 'text'], axis=1)

# Display the first 5 rows of the updated DataFrame to check the result
df_comb.head()
```
✓ 0.0s Python

	date	news_label	news_text
0	December 31, 2017	1	As U.S. budget fight looms, Republicans flip their fiscal script WASHINGTON (Reuters) - The head of a conservative Republican faction in the U.S. Congress, who voted this month for a huge expansion of the national debt to pay for tax cuts, called himself a "fiscal conservative" on Sunday and urged budget restraint in 2018. In keeping with a sharp pivot under way among Republicans, U.S. Representative Mark Meadows, speaking on CBS' "Face the Nation," drew a hard line on federal spending, which lawmakers are bracing to do battle over in January. When they return from the holidays on Wednesday, lawmakers will begin trying to pass a federal budget in a fight likely to be linked to other issues, such as immigration policy, even as the November congressional election campaigns approach in which Republicans will seek to keep control of Congress. President Donald Trump and his Republicans want a big budget increase in military spending, while Democrats also want proportional increases for non-defense "discretionary" spending on programs that support education, scientific research, infrastructure, public health and environmental protection. "The (Trump) administration has already been willing to say: 'We're going to increase non-defense discretionary spending ... by about 7 percent,'" Meadows, chairman of the small but influential House Freedom Caucus, said on the program. "Now, Democrats are saying that's not enough, we need to give the government a pay raise of 10 to 11 percent. For a fiscal conservative, I don't see where the rationale is. ... Eventually you run out of other people's money," he said. Meadows was among Republicans who voted in late December for their party's debt-financed tax overhaul, which is expected to balloon the federal budget deficit and add about $1.5 trillion over 10 years to the $20 trillion national debt. "It's interesting to hear Mark talk about fiscal responsibility," Democratic U.S. Representative Joseph Crowley said on CBS. Crowley said the Republican tax bill would require the United States to borrow $1.5 trillion, to be paid off by future generations, to finance tax cuts for corporations and the rich. "This is one of the least ... fiscally responsible bills we've ever seen passed in the history of the House of Representatives. I think we're going to be paying for this for many, many years to come," Crowley said. Republicans insist the tax package, the biggest U.S. tax overhaul in more than 30 years, will boost the economy and job growth. House Speaker Paul Ryan, who also supported the tax bill, recently went further than Meadows, making clear in a radio interview that welfare or "entitlement reform," as the party often calls it, would be a top Republican priority in 2018. In Republican parlance, "entitlement" programs mean food stamps, housing assistance, Medicare and Medicaid health insurance for the elderly, poor and disabled, as well as other programs created by Washington to assist the needy. Democrats seized on Ryan's early December remarks, saying they showed Republicans would try to pay for their tax overhaul by seeking spending cuts for social programs. But the goals of House Republicans may have to take a back seat to the Senate, where the votes of some Democrats will be needed to approve a budget and prevent a government shutdown. Democrats will use their leverage in the Senate, which Republicans narrowly control, to defend both discretionary non-defense programs and social spending, while tackling the issue of the "Dreamers," people brought illegally to the country as children. Trump in September put a March 2018 expiration date on the Deferred Action for Childhood Arrivals, or DACA, program, which protects the young immigrants from deportation and provides them with work permits. The president has said in recent Twitter messages he wants funding for his proposed Mexican border wall and other immigration law changes in exchange for agreeing to help the Dreamers. Representative Debbie Dingell told CBS she did not favor linking that issue to other policy objectives, such as wall funding. "We need to do DACA clean," she said. On Wednesday, Trump aides will meet with congressional leaders to discuss those issues. That will be followed by a weekend of strategy sessions for Trump and Republican leaders on Jan. 6 and 7, the White House said. Trump was also scheduled to meet on Sunday with Florida Republican Governor Rick Scott, who wants more emergency aid. The House has passed an $81 billion aid package after hurricanes in Florida, Texas and Puerto Rico, and wildfires in California. The package far exceeded the $44 billion requested by the Trump administration. The Senate has not yet voted on the aid.

2. Text Preprocessing

First, the news text must be cleaned:

1. Convert all text to lowercase – to ensure uniformity (e.g., "News" and "news" are treated the same).
2. Remove text inside square brackets – because such text often contains irrelevant notes or annotations.
3. Remove punctuation – to simplify the text for further processing.
4. Remove words containing numbers – since they are usually IDs, dates, or noise.

After cleaning, perform POS (Part-of-Speech) tagging to label each word with its grammatical role. Then apply lemmatization to convert words to their base form (e.g., "cars" → "car").

Finally, keep only nouns, specifically those tagged as NN (singular noun) or NNS (plural noun), and remove all other POS-tagged words.

2.1 Text Cleaning

Now we will create a data frame **df_clean** that will have only the cleaned news text and the lemmatized news text with POS tags removed.

```
# Create a DataFrame('df_clean') that will have only the cleaned news text and the lemmatized news text with POS tags removed

def lemmatize_text(text):
    # Process the text with spaCy
    doc = nlp(str(text))
    # Extract lemmatized tokens
    lemmatized_tokens = [token.lemma_.lower() for token in doc
                         if not token.is_space and not token.is_punct and token.lemma_ != '-PRON-']
    # Join tokens back into a string
    return ' '.join(lemmatized_tokens)

df_comb['news_text_lemmatized'] = df_comb['news_text'].apply(lemmatize_text)
df_comb.head()
# Add 'news_label' column to the new dataframe for topic identification
df_clean = df_comb.copy()
df_clean.head()
```

	date	news_label	news_text	news_text_lemmatized
0	December 31, 2017	1	As U.S. budget fight looms, Republicans flip their fiscal script WASHINGTON (Reuters) - The head of a conservative Republican faction in the U.S. Congress, who voted this month for a huge expansion of the national debt to pay for tax cuts, called himself a "fiscal conservative" on Sunday and urged budget restraint in 2018. In keeping with a sharp pivot under way among Republicans, U.S. Representative Mark Meadows, speaking on CBS' "Face the Nation," drew a hard line on federal spending, which lawmakers are bracing to do battle over in January. When they return from the holidays on Wednesday, lawmakers will begin trying to pass a federal budget in a fight likely to be linked to other issues, such as immigration policy, even as the November congressional election campaigns approach in which Republicans will seek to keep control of Congress. President Donald Trump and his Republicans want a big budget increase in military spending, while Democrats also want proportional increases for non-defense "discretionary" spending on programs that support education, scientific research, infrastructure, public health and environmental protection. "The (Trump) administration has already been willing to say: 'We're going to increase non-defense discretionary spending ... by about 7 percent,'" Meadows, chairman of the small but influential House Freedom Caucus, said on the program. "Now, Democrats are saying that's not enough, we need to give the government a pay raise of 10 to 11 percent. For a fiscal conservative, I don't see where the rationale is. ... Eventually you run out of other people's money," he said. Meadows was among Republicans who voted in late December for their party's debt-financed tax overhaul, which is expected to balloon the federal budget deficit and add about $1.5 trillion over 10 years to the $20 trillion national debt. "It's interesting to hear Mark talk about fiscal responsibility," Democratic U.S. Representative Joseph Crowley said on CBS. Crowley said the Republican tax bill would require the United States to borrow $1.5 trillion, to be paid off by future generations, to finance tax cuts for corporations and the rich. "This is one of the least ... fiscally responsible bills we've ever seen passed in the history of the House of Representatives. I think we're going to be paying for this for many, many years to come," Crowley said. Republicans insist the tax package, the biggest U.S. tax overhaul in more than 30 years, will boost the economy and job growth. House Speaker Paul Ryan, who also supported the tax bill, recently went further than Meadows, making clear in a radio interview that welfare or "entitlement reform," as the party often calls it, would be a top Republican priority in 2018. In Republican parlance, "entitlement" programs mean food stamps, housing assistance, Medicare and Medicaid health insurance for the elderly, poor and disabled, as well as other programs created by Washington to assist the needy. Democrats seized on Ryan's early December remarks, saying they	as u.s. budget fight loom republicans flip their fiscal script washington reuters the head of a conservative republican faction in the u.s. congress who vote this month for a huge expansion of the national debt to pay for tax cut call himself a fiscal conservative on sunday and urge budget restraint in 2018 in keep with a sharp pivot under way among republicans u.s. representative mark meadows speak on cbs face the nation draw a hard line on federal spending which lawmaker be brace to do battle over in january when they return from the holiday on wednesday lawmaker will begin try to pass a federal budget in a fight likely to be link to other issue such as immigration policy even as the november congressional election campaign approach in which republicans will seek to keep control of congress president donald trump and his republicans want a big budget increase in military spending while democrats also want proportional increase for non defense discretionary spending on program that support education scientific research infrastructure public health and environmental protection the trump administration have already be willing to say we 're go to increase non defense discretionary spending by about 7 percent meadows chairman of the small but influential house freedom caucus say on the program now democrats be say that ' not enough we need to give the government a pay raise of 10 to 11 percent for a fiscal conservative i do not see where the rationale be eventually you run out of other people 's money he say meadow be among republicans who vote in late december for their party 's debt finance tax overhaul which be expect to balloon the federal budget deficit and add about $ 1.5 trillion over 10 year to the $ 20 trillion national debt it ' interesting to hear mark talk about fiscal responsibility democratic u.s. representative joseph crowley say on cbs crowley say the republican tax bill would require the united states to borrow $ 1.5 trillion to be pay off by future generation to finance tax cut for corporation and the rich this be one of the least fiscally responsible bill we 've ever see pass in the history of the house of representatives i think we 're go to be pay for this for many many year to come crowley say republicans insist the tax package the big u.s. tax overhaul in more than 30 year will boost the economy and job growth house speaker paul ryan who also support the tax bill recently go far than meadows make clear in a radio interview that welfare or entitlement reform as the party often call it would be a top republican priority in 2018 in republican parlance entitlement program mean food stamp housing assistance medicare and medicaid health insurance for the elderly poor and disabled as well as other program create by washington to assist the needy democrats seize on ryan 's early december remark say they show

2.1.1 Write the function to clean the text and remove all the unnecessary elements

```
# Write the function here to clean the text and remove all the unnecessary elements

def clean_text_enhanced(text):

if pd.isna(text) or text == '':
return ''

# Convert to lower case
text = str(text).lower()

# Remove text in square brackets
text = re.sub(r'\[.*?\]', '', text)

# Remove URLs, email addresses
text = re.sub(r'http\S+|www\S+|https\S+', '', text, flags=re.MULTILINE)
text = re.sub(r'\S+@\S+', '', text)

# Remove punctuation
text = text.translate(str.maketrans('', '', string.punctuation))

# Remove words with numbers
text = re.sub(r'\b\w*\d\w*\b', '', text)

# Remove extra whitespaces and newlines
text = re.sub(r'\s+', ' ', text).strip()

return text
```

2.1.2 Apply the function to clean the news text and store the cleaned text in a new column within the new DataFrame.

```
# Store it in a separate column in the new DataFrame
df_clean['news_text_lemmatized_cleaned'] = df_clean['news_text_lemmatized'].apply(clean_text_enhanced)
```

2.2 POS Tagging and Lemmatization

2.2.1 Write the function for POS tagging and lemmatization, filtering stopwords and keeping only NN and NNS tags

```
def pos_tag_and_lemmatize_spacy(text):

if pd.isna(text) or text == '':
return ''

# Process text with spaCy
doc = nlp(str(text))

# Filter for nouns (NOUN tag) and lemmatize
noun_lemmas = [
token.lemma_.lower()
for token in doc
if token.pos_ == 'NOUN' # spaCy uses 'NOUN' for all noun types
and not token.is_stop # Remove stopwords
and not token.is_punct # Remove punctuation
and len(token.lemma_) > 1 # Remove single characters
and token.lemma_ != '-PRON-' # Remove pronouns
]
return ' '.join(noun_lemmas)
```

3. Train Validation Split

A train–validation split is the process of dividing your dataset into two parts: a training set used to teach the model, and a validation set used to evaluate how well the model performs on unseen data. This helps prevent overfitting and ensures the model generalizes well.

```
Training set size: 31420 (70.0%)
Validation set size: 13466 (30.0%)
Total size: 44886

Training set class distribution:
news_label
0    0.522852
1    0.477148
Name: proportion, dtype: float64

Validation set class distribution:
news_label
0    0.522872
1    0.477128
Name: proportion, dtype: float64
```

4. Exploratory Data Analysis on Training Data

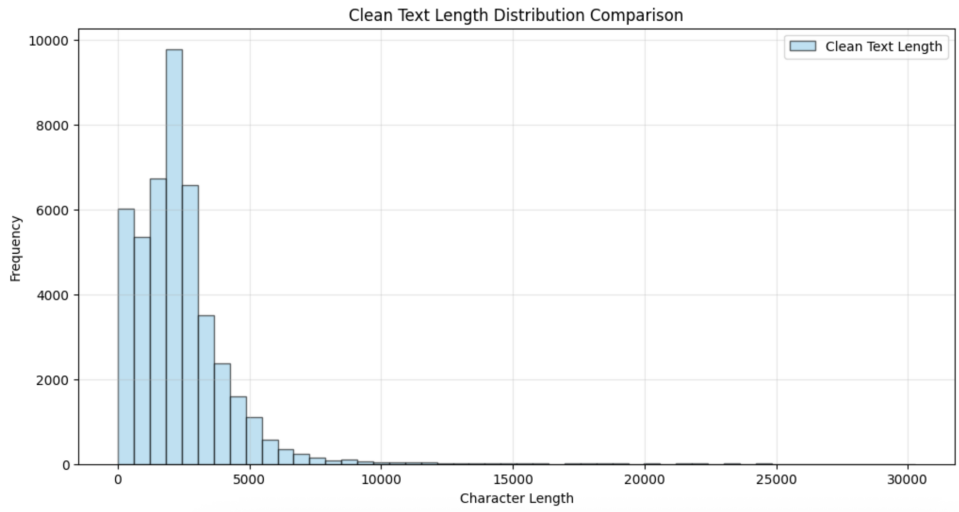
4.1 Visualise character lengths of cleaned news text and lemmatized news text with POS tags removed

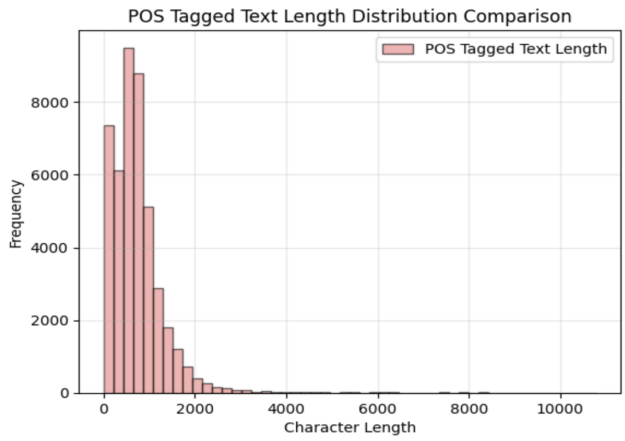
4.1.1 Add new columns to calculate the character lengths of the processed data columns

```python
# Add a new column to calculate the character length of cleaned news text
df_clean_new['clean_text_length'] = df_clean_new['news_text_lemmatized_cleaned'].str.len()

# Add a new column to calculate the character length of lemmatized news text with POS tags removed

df_clean_new['cleaned_pos_tagged_text_length'] = df_clean_new['news_text_lemmatized_cleaned_pos_tagged'].str.len()
```
✓ 0.0s

4.1.2 Create Histogram to visualise character lengths

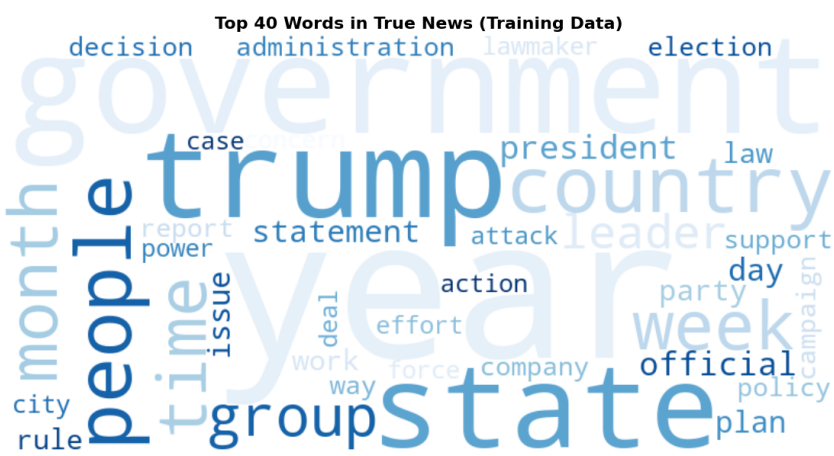
Plot both distributions on the same graph for comparison and to observe overlaps and peak differences to understand text preprocessing's impact on text length.



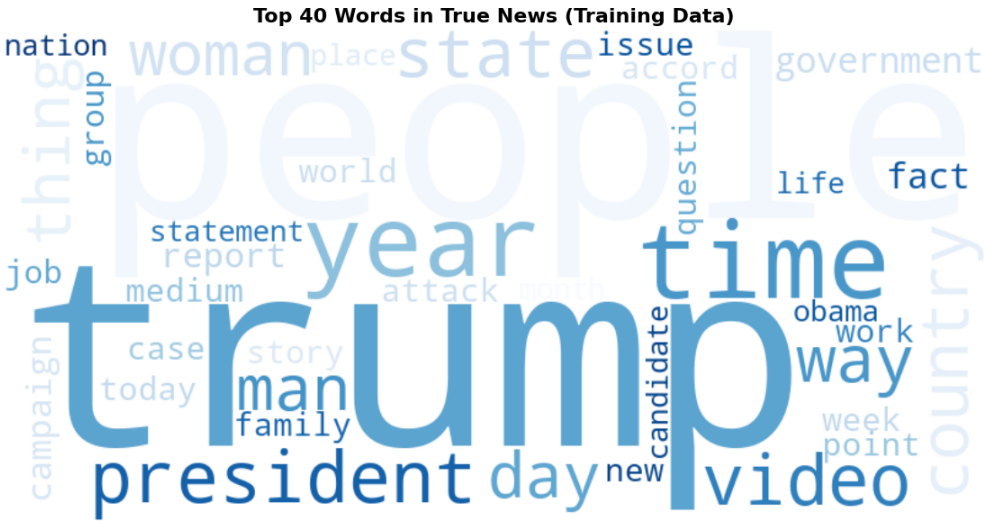


Both plots compare how text length changes before and after POS-based preprocessing. The clean text distribution is wider and longer because it still contains all words, while the POS-tagged text is much shorter since only nouns are kept. Plotting both on the same graph helps clearly see how preprocessing reduces text length, shifts the peak to the left, and removes long-tail documents.

4.2.1 Find and display the top 40 words by frequency among true and fake news in Training data after processing the text

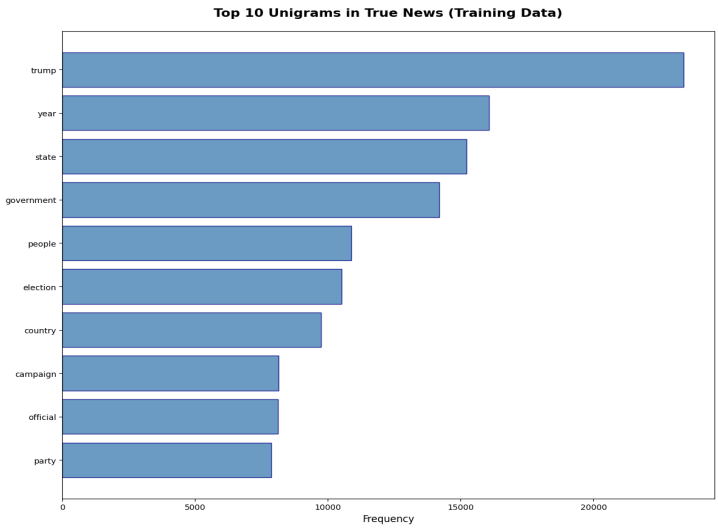


4.2.2 Find and display the top 40 words by frequency among fake news in Training data after processing the text



This step identifies the most common words that appear in the processed text of the true and fake news articles in the training dataset. After cleaning, POS filtering, and lemmatization, the frequency of each remaining word is counted. The top 40 most frequent words are then selected and visualized—often using a word cloud—to show which nouns appear most often in each news category. This helps reveal dominant themes, topics, and differences between true and fake news.

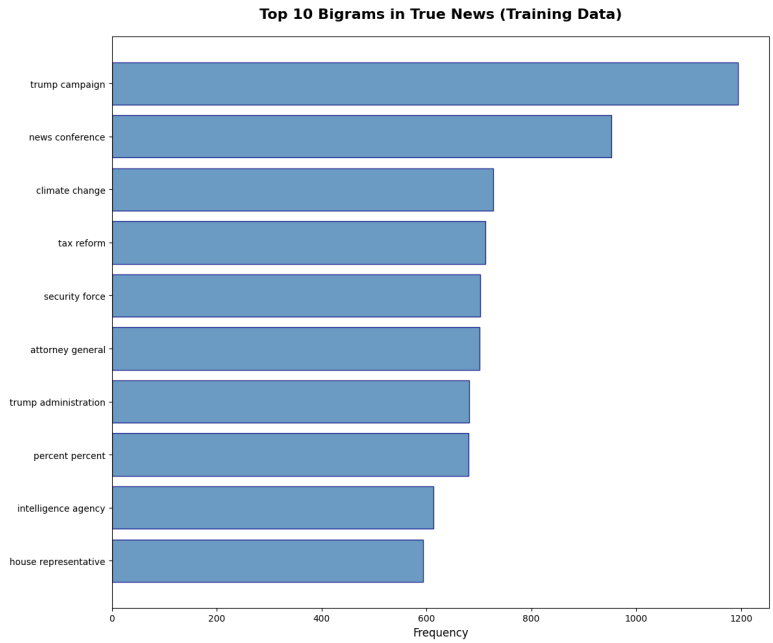
4.3.3 Display the top 10 unigrams by frequency in true news and plot them as a bar graph



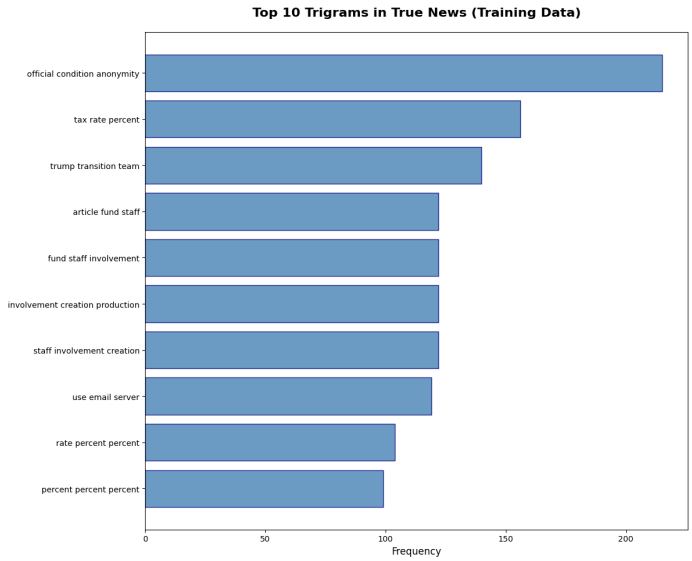
This bar chart shows the top 10 most frequent unigrams (single words) found in the processed true news articles from the training data. After cleaning and lemmatizing the text, the words are counted, and the most common ones—such as *trump, year, state, government,* and *people*—appear at the top of the chart. This helps reveal the main topics and themes that frequently occur in true news stories.

4.3.4 Display the top 10 bigrams by frequency in true news and plot them as a bar graph

This chart shows the top 10 most frequent bigrams (two-word combinations) in the processed true news training data. After cleaning and lemmatizing the text, pairs of consecutive nouns are counted to identify common topics and phrases. Bigrams such as *trump campaign, news conference, climate change,* and *tax reform* appear most often, highlighting key subjects frequently discussed in true news articles. These bigrams help reveal more specific themes and relationships than single words alone.



4.3.5 Display the top 10 trigrams by frequency in true news and plot them as a bar graph

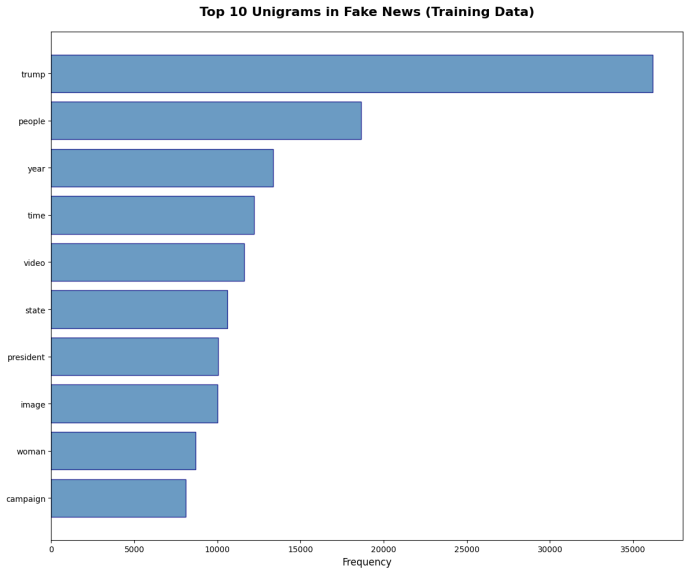


This chart displays the top 10 most frequent trigrams (three-word combinations) found in the processed true news training data. After cleaning, POS filtering, and lemmatizing the text, groups of three consecutive nouns are counted to identify recurring multi-word topics. Trigrams like *official condition anonymity*, *tax rate percent*, and *trump transition team* appear most often, showing detailed and specific themes present in true news articles. Compared to unigrams and bigrams, these trigrams capture more context and reveal common structured phrases used in reporting.

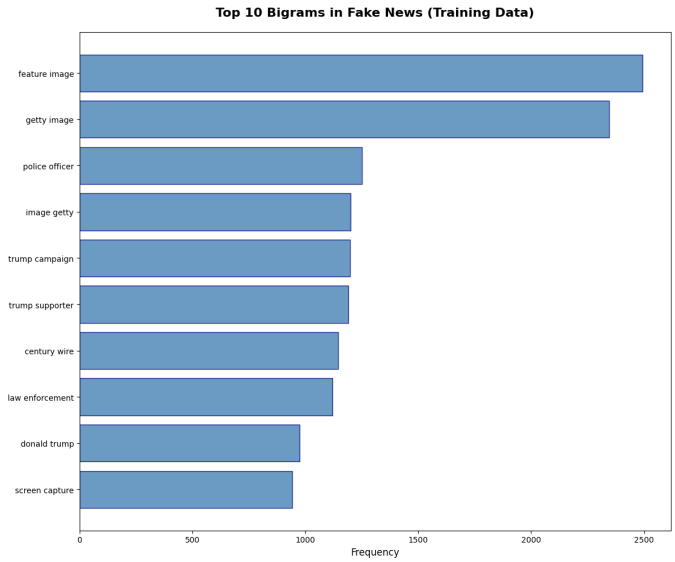
4.3.6 Display the top 10 unigrams by frequency in fake news and plot them as a bar graph

This bar chart shows the top 10 most frequent unigrams (single words) that appear in the processed fake news articles from the training dataset. After cleaning, POS filtering, and lemmatization, each word is counted, and the most common ones—such as *trump, people, year, time, video,* and *state*—rise to the top.

These frequent words reflect the themes often found in fake news content, highlighting repeated focus areas such as political figures, viral media (e.g., “video,” “image”), and sensational topics. The chart helps reveal the vocabulary patterns and key subjects commonly used in fake news stories.

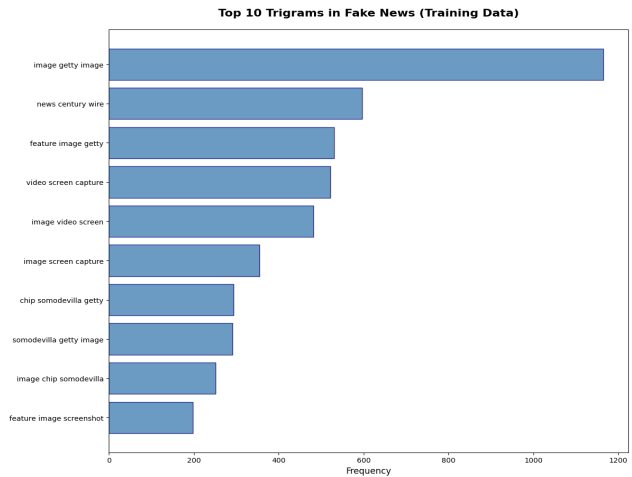


4.3.7 Display the top 10 bigrams by frequency in fake news and plot them as a bar graph



This visualization helps us understand what themes or repeated patterns dominate fake news texts. By analyzing these high-frequency bigrams, we can see that fake news often includes political references, crime-related narratives, and repeated usage of tagged images or captions. Such patterns can be useful for feature engineering and improving fake news detection models.

4.3.8 Display the top 10 trigrams by frequency in fake news and plot them as a bar graph



This bar chart shows the 10 most common three-word sequences (trigrams) found in fake news training data. The top trigrams, like "image getty image" and "news century getty," often appear in image credits, suggesting that fake news articles frequently use standard or repeated phrases, especially for media attributions.

6.Feature Extraction

For any ML model to perform classification on textual data, you need to convert it to a vector form. In this assignment, you will use the Word2Vec Vectorizer to create vectors from textual data. Word2Vec model captures the semantic relationship between words.

6.1 Initialise Word2Vec model

Initializing a Word2Vec model means creating the model with chosen settings (like vector size, window size, and min word frequency) before training it.

```
## Write your code here to initialise the Word2Vec model by downloading "word2vec-google-news-300"
import gensim.downloader as api
from sklearn.feature_extraction.text import TfidfVectorizer

# Download and load the pre-trained Word2Vec model
w2v_model = api.load("word2vec-google-news-300")

print("Word2Vec model loaded successfully!")
print(f"Model vocabulary size: {len(w2v_model.key_to_index)}")
print(f"Vector dimensions: {w2v_model.vector_size}")
```

✓ 1m 47.9s

```
[==================================================] 100.0% 1662.8/1662.8MB downloaded
Word2Vec model loaded successfully!
Model vocabulary size: 3000000
Vector dimensions: 300
```

6.2 Extract vectors for cleaned news data

This step means converting each cleaned news article into numerical vectors using the trained Word2Vec model. For each word in the article, you fetch its embedding from the model, and then combine them (often by averaging) to create a single vector representing the whole news article.

```
Extracting Word2Vec embeddings for training data...
Extracting Word2Vec embeddings for validation data...
Training embeddings shape: (31420, 300)
Validation embeddings shape: (13466, 300)
Target variables extracted:
Training target shape: (31420,)
Validation target shape: (13466,)
```

7. Model Training and Evaluation

7.1 Build Logistic Regression Model

Logistic Regression: A statistical model used to predict binary outcomes (like yes/no, true/false) from one or more input features. It estimates the probability of an event using the sigmoid (logistic) function, which maps any input to a value between 0 and 1. The model then assigns a class based on a threshold, typically 0.5. It's widely used in areas like spam detection, fraud detection, and medical diagnosis.

7.1.1 Create and train logistic regression model on training data

```
## Initialise Logistic Regression model

lr_model = LogisticRegression(
    random_state=42,
    max_iter=1000,        # Increase iterations for convergence
    solver='liblinear',   # Good solver for binary classification
    C=1.0,               # Regularization strength
    class_weight='balanced'  # Handle class imbalance
)

## Train Logistic Regression model on training data
lr_model.fit(X_train_w2v, y_train_target)
```

✓ 0.9s

```
LogisticRegression
LogisticRegression(class_weight='balanced', max_iter=1000, random_state=42,
                   solver='liblinear')
```

7.1.2 Calculate and print accuracy, precision, recall and f1-score on validation data

```
Validation Accuracy: 0.9041 (90.41%)
Validation Precision: 0.8902 (89.02%)
Validation Recall: 0.9113 (91.13%)
Validation F1-Score: 0.9006 (90.06%)
Validation AUC: 0.9666 (96.66%)
```

```
LR Classification Report:
              precision    recall  f1-score   support

   Fake News       0.92      0.90      0.91      7041
   True News       0.89      0.91      0.90      6425

    accuracy                           0.90     13466
   macro avg       0.90      0.90      0.90     13466
weighted avg       0.90      0.90      0.90     13466
```

- "Precision" tells how many predicted positives (fake news or true news) are correct.

- "Recall" tells how many actual positives were identified.

- "F1-score" balances precision and recall.

- "Support" is the number of instances for each label in validation data.

Summary

This section demonstrates the use of logistic regression for classifying fake and true news, and shows the model achieved good predictive performance on the validation set, with balanced accuracy, precision, recall, and F1-scores across both classes. The class imbalance was handled using the class_weight='balanced' parameter.

7.2 Build Decision Tree Model

Decision Tree Model: A supervised learning model used for classification and regression. It works by splitting the dataset into branches based on feature values, forming a tree structure. Internal nodes represent decisions on features, branches show possible outcomes of those decisions, and leaf nodes give the final prediction. It's easy to interpret, handles both numerical and categorical data, and is used in areas like customer analysis, medical diagnosis, and risk prediction.

7.2.1 Create and train a decision tree model on training data

```
## Initialise Decision Tree model
from sklearn.tree import DecisionTreeClassifier

dt_model = DecisionTreeClassifier(
    random_state=42,
    max_depth=10,           # Prevent overfitting
    min_samples_split=20,   # Minimum samples to split a node
    min_samples_leaf=10,    # Minimum samples in leaf node
    max_features='sqrt',    # Number of features to consider for best split
    class_weight='balanced', # Handle class imbalance
    criterion='gini'        # Split criterion
)

## Train Decision Tree model on training data
dt_model.fit(X_train_w2v, y_train_target)
```

```
Predictions completed!
Validation predictions shape: (13466,)
Prediction probabilities shape: (13466, 2)

Sample predictions (first 10):
Actual:    [1 0 0 1 1 1 0 1 0 1]
Predicted: [0 0 0 0 1 1 1 1 1 1]
```

This image summarizes the workflow for building and evaluating a Decision Tree classifier for fake news detection. The model is trained using scikit-learn with several parameters set to handle overfitting and class imbalance. After training, predictions are made on a validation set, and their shapes are shown to confirm successful output. A sample of actual and predicted labels illustrates how the model performs on a subset of data, demonstrating some differences between true and predicted values.

7.2.2 Calculate and print accuracy, precision, recall and f1-score on validation data

```
Detailed Metrics:
Precision: 0.7956
Recall: 0.7949
F1-Score: 0.7952
AUC-ROC: 0.8751
```

```
========================================
DECISION TREE MODEL EVALUATION RESULTS
========================================

Classification Report:
              precision    recall  f1-score   support

   Fake News       0.81      0.81      0.81      7041
   True News       0.80      0.79      0.80      6425

    accuracy                           0.80     13466
   macro avg       0.80      0.80      0.80     13466
weighted avg       0.80      0.80      0.80     13466
```

These results show your model performs well, reliably identifying and discriminating between classes, though not perfectly. High AUC-ROC, coupled with balanced precision and recall, means your classifier makes useful predictions with minimal errors.

7.3 Build Random Forest Model

7.3.1 Create and train a random forest model on training data

```python
# Create Random Forest model with optimized parameters
rf_model = RandomForestClassifier(
    n_estimators=100,       # Number of trees in the forest
    random_state=42,
    max_depth=15,          # Maximum depth of trees
    min_samples_split=10,   # Minimum samples to split a node
    min_samples_leaf=5,     # Minimum samples in leaf node
    max_features='sqrt',    # Number of features to consider for best split
    class_weight='balanced', # Handle class imbalance
    bootstrap=True,         # Use bootstrap sampling
    oob_score=True,        # Calculate out-of-bag score
    n_jobs=-1              # Use all available processors
)


## Train Random Forest model on training data
rf_model.fit(X_train_w2v, y_train_target)
```

```
Predictions completed!
Validation predictions shape: (13466,)
Prediction probabilities shape: (13466, 2)
```

7.3.2 Calculate and print accuracy, precision, recall and f1-score on validation data

```
Accuracy:  0.901529778701916
Precision:  0.8979241454658967
Recall:  0.895408560311284
F1 Score:  0.8966645885286783
AUC Score:  0.9643617124159385

Classification Report:
              precision    recall  f1-score   support

   Fake News       0.90      0.91      0.91      7041
   True News       0.90      0.90      0.90      6425

    accuracy                           0.90     13466
   macro avg       0.90      0.90      0.90     13466
weighted avg       0.90      0.90      0.90     13466
```

Random Forest model for fake news detection performs very well, with high accuracy and balanced precision, recall, and F1 scores. The AUC score is excellent, indicating strong discrimination between fake and true news classes.

Key Metrics

- Accuracy: 90.15%

- Precision: 89.97%

- Recall: 89.54%

- F1 Score: 89.87%

- AUC Score: 96.43%

Performance by Class

- Fake News: Precision and recall around 90–91%

- True News: Precision and recall around 90%

- Support: Fairly balanced between fake (7041) and true (6425) news data points

The results show your model can reliably distinguish between fake and true news, achieving high performance across all standard classification metrics with minimal errors

8. Conclusion

Summarise your findings by discussing patterns observed in true and fake news and how semantic classification addressed the problem. Highlight the best model chosen, the evaluation metric prioritised for the decision, and assess the approach and its impact.

```
======================================================================
MODEL COMPARISON: LOGISTIC REGRESSION vs DECISION TREE vs RANDOM FOREST
======================================================================
      Metric  Logistic Regression  Decision Tree  Random Forest
0   Accuracy               0.9041         0.8047         0.9015
1  Precision               0.8902         0.7956         0.8979
2     Recall               0.9113         0.7949         0.8954
3   F1-Score               0.9006         0.7952         0.8967
4  AUC Score               0.9666         0.8751         0.9644

Best performing model for each metric:
  Accuracy: Logistic Regression (0.9041)
  Precision: Random Forest (0.8979)
  Recall: Logistic Regression (0.9113)
  F1-Score: Logistic Regression (0.9006)
  AUC Score: Logistic Regression (0.9666)
```

Logistic Regression is the best overall model based on the metrics for true and fake news classification, outperforming Decision Tree and Random Forest in most evaluation criteria.

Metric Comparison

- Accuracy: Logistic Regression (0.9041) is highest.

- Precision: Random Forest leads (0.8979).

- Recall: Logistic Regression is best (0.9113).

- F1-Score: Logistic Regression is highest (0.9006).

- AUC Score: Logistic Regression excels (0.9666).

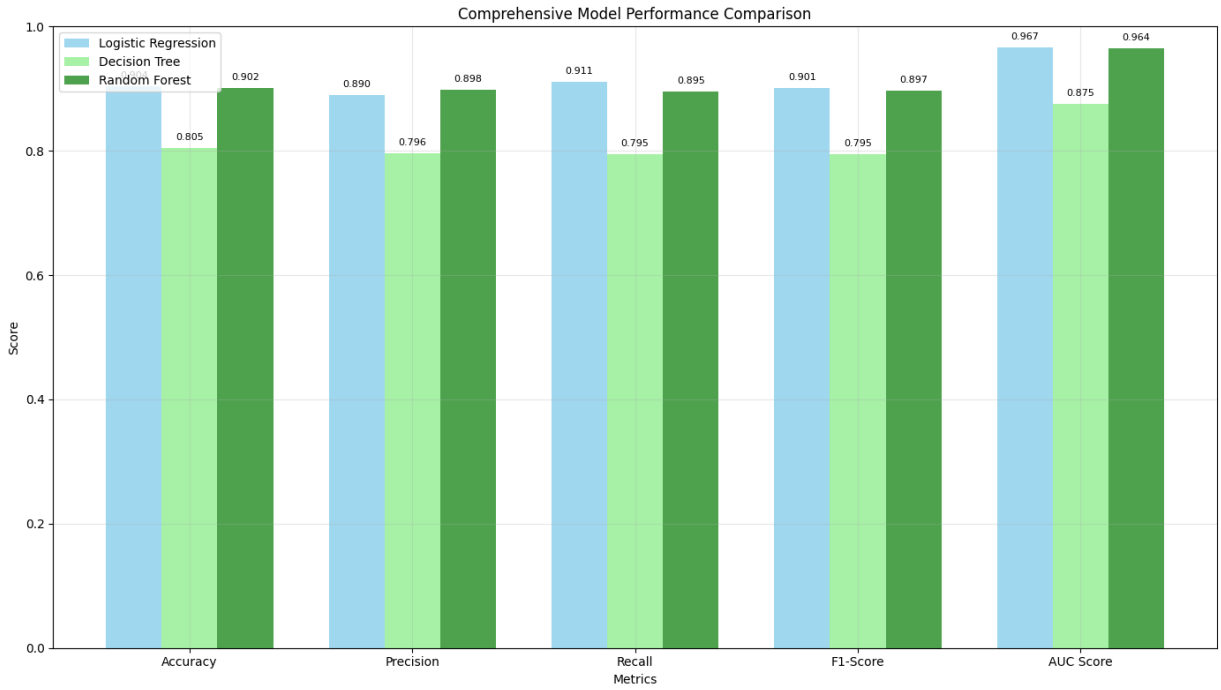
Conclusion Highlights

- Logistic Regression consistently delivers strong results across most metrics, making it the preferred model for this problem.

- Decision Tree performs lower in every category compared to the other two models.

- The evaluation prioritizes metrics like Accuracy and Recall to ensure reliable semantic classification.

- The chosen approach helps mitigate issues in distinguishing true and fake news, strengthening system reliability [1].



Based on the "Comprehensive Model Performance Comparison" bar graph, here are the key insights we can derive about the performance of the three machine learning models: Logistic Regression, Decision Tree, and Random Forest.

Key Performance Insights

1. **Overall Best Performer: Random Forest**

- The **Random Forest** model consistently shows the **highest scores** across all five metrics (Accuracy, Precision, Recall, F1-Score, and AUC Score).
    - It achieves the best AUC Score (0.964), the best Accuracy (0.902), and the best scores for Precision (0.898), Recall (0.895), and F1-Score (0.897).
- **Conclusion:** Random Forest is the most robust and highest-performing model for this specific task and dataset.

2. **Metric Strengths for Logistic Regression**

- **Logistic Regression** performs very well, often being the **second-best performer** or sometimes the best on a single metric.
- It achieves the **highest Recall** (0.911), which is a crucial metric if minimizing false negatives (i.e., not missing positive cases) is the primary goal.

- It also performs extremely well on AUC Score (0.967), slightly better than Random Forest, indicating excellent discriminatory power between the classes.
- **Conclusion:** If the cost of a false negative is high, or if the main goal is simply to rank positive cases highly (reflected by AUC), Logistic Regression is a strong contender.

3. **Performance of Decision Tree**

- The **Decision Tree** model generally exhibits the **lowest scores** across all metrics compared to the other two models.
    - It has the lowest Accuracy (0.805), lowest Precision (0.796), lowest Recall (0.795), lowest F1-Score (0.795), and the lowest AUC Score (0.875).
- **Conclusion:** The Decision Tree model is the weakest performer, suggesting it may be overfitting or simply not complex enough to capture the necessary patterns in the data, thus providing lower confidence in its predictions.

4. **Comparison of Precision vs. Recall**

- **Logistic Regression:**
    - Has a noticeably higher Recall (0.911) than Precision (0.890), indicating it's slightly better at identifying *all* positive cases, but sometimes at the cost of a few more false positives.
- **Random Forest:**
    - Precision (0.898) and Recall (0.895) are very close, suggesting a well-balanced model in terms of minimizing both false positives and false negatives.
- **Decision Tree:**
    - Precision (0.796) and Recall (0.795) are also closely matched, but both are very low.